Task Sheet No. 1:
ACTION RESEARCH DATA BANK

INSTRUCTION:

As a group emerging teacher-researcher, you are task to develop an **Action Research Data Bank** based on actual observations and experiences during your Field Study and from the latest review or related literatures (RRL)/published action research. You will identify, document, and organize potential **classroom-based and school-based problems or concerns** that may serve as possible topics for future action research. It may be related to different classroom situations, teaching practices, learner behaviors, instructional strategies, assessment practices, and school-related concerns that you encountered. Some of these situations may indicate a gap, difficulty, recurring problem, or opportunity for improvement that can be addressed through action research. Your observations must be based on **actual Field Study experiences** and should not be based solely on assumptions or hypothetical situations. Your output will be submitted to your Field Study Instructor/Professor and may be used as a reference when developing a future Classroom-Based Action Research (CBAR) proposal.

RUBRIC FOR GRADING:

CRITERIA	EXCELLENT	PROFICIENT	DEVELOPING	BEGINNING
Thematic Area and Context (20%)	Clearly identifies an appropriate thematic area and provides a specific, relevant, and well-contextualized description of the classroom/school situation where the concern was observed.	Identifies an appropriate thematic area and provides a clear description of the relevant classroom/school context.	Identifies a thematic area and provides a general context, but important details or connections are lacking.	The thematic area or context is unclear, inappropriate, not connected or missing.
Identification of Research Concerns/Problem (25%)	Identifies 10 or more specific, relevant, and clearly researchable concerns/problem based on actual Field Study experiences and RRL	Identifies at least 8-10 relevant concerns/problem and RRL, with most being specific and potentiality.	Identifies at least 5-7 concerns and/or RRL and several concerns/problem are broad, unclear, or difficult to research.	Identifies less than 5 concerns/problem and RRL, which are mostly vague, unrelated, or not researchable.
Identification and Analysis of the Problem, Solution, and Outcome (25%)	Clearly identifies a specific and observable educational concern and provides thoughtful analysis of its possible causes, solution, and effects.	Clearly identifies the concern and provides an adequate explanation of its possible causes, solution, or effects.	Identifies the concern, solution, and outcome but provides limited analysis or relies mainly on description.	Concern, solution, and outcome is vague, unsupported, or difficult to identify.
Alignment of Components (20%)	Demonstrates strong and logical alignment among the thematic area, context, identified concern, and proposed solution/action plan.	Shows clear alignment among most of the required components but not all.	Shows partial alignment, with some inconsistencies among the components.	Components are disconnected or demonstrate little logical relationship.
Organization and Completeness (10%)	All required components are complete, logically organized, coherent, and presented clearly.	Required components are complete and generally well organized.	Some components are incomplete or inconsistently organized.	Several required components are missing or poorly organized.

Suggested Proposal Title	Improving the Efficiency of Student Attendance and Exit and Return Logging Through a QR-Code-Based Mobile Application in Pres. Diosdado Macapagal High School
Research Agenda or Thematic Area/s	For Teaching and Learning
Research Focus	Learners
Name of Proponents	Arellano, Ailyn P. – Electronics Technology Duran, John Rex B. – ICT Computer Programming Porteria, Kimberlie Crissel F. – ICT Computer Programming
Name of Adviser	Under Consultation

Activity 1: Write your answer in the space provided.

Action Research Titles	Author/s (APA Format)	Thematic Area/ Context	Summary (Problem, Action Plan, and Outcome)
Example: Differentiated Instruction in teaching TLE for Grade 8 Classes		Thematic Area: Teaching and Learning Focus: Instruction Context: Describe briefly	Problem: Action/Solution: Outcome:
1. Assessing the Impact of QR Code Scanning on Attendance Monitoring and Punctuality among Grade 12 Learners in Section Athena of Ormoc City Regional Sports Academy	Pabular, E. J. Y. (2025).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Classroom Management. **Context:** The study was conducted among Grade 12 learners at Ormoc City Regional Sports Academy. It examined the use of QR code scanning as an alternative to manual attendance monitoring, focusing on improving punctuality, attendance-recording accuracy, and monitoring efficiency.	**Problem:** Traditional manual attendance monitoring was less efficient and accurate than the proposed digital approach. The study also addressed the need to improve learners' punctuality and establish a more effective attendance-monitoring process. **Action/Solution:** The researcher implemented a QR-code-based attendance monitoring system. Learners in the experimental group (Grade 12 Athena, 40 learners) used QR code scanning, while the control group (Grade 12 Janus, 40

			learners) continued using traditional attendance methods. The implementation lasted 25 days. Attendance records and a post-implementation questionnaire were used to evaluate the system. **Outcome:** The experimental group demonstrated higher punctuality than the control group. Learners generally found the QR code system convenient, user-friendly, and efficient. The study also reported 95% accuracy for QR-based attendance compared with 80% for traditional attendance. Learners reported that attendance tracking became easier and more time-efficient.
2. Enhancing Student Safety and Accountability through a QR Code-Based Gate Monitoring System for Patao National High School (PNHS)	Ybañez, R. M., Botilla, J. E., Pacaldo, E. I., Jr., Fernandez, R. M., Jr., & Alegre, K. B. S. (2026).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and School Management. **Context:** The study was conducted at Patao National High School, a public secondary school in the Philippines with approximately 4,000 students from Grades 7–12. It focused on improving student safety, accountability, and monitoring of school gate entry and exit through a QR-code-based system.	**Problem:** Traditional manual gate monitoring was time-consuming, prone to human error, and unable to provide immediate updates to parents regarding students' entry and exit. Unauthorized departures, truancy, and limited parental awareness were also identified as concerns. **Action/Solution:** The researchers developed and implemented a QR-code-based gate monitoring system. Students scanned their unique QR codes upon entering or leaving the

			school premises. School guards verified student identities, while the system automatically recorded timestamped transactions in a centralized database and sent email notifications to parents or guardians. **Outcome:** The system demonstrated improved monitoring accuracy, reduced manual recording errors, and faster processing, with verification taking under five seconds during peak tests. The evaluation obtained an overall mean of 4.36 (Strongly Agree) for system functionality and 4.65 (Strongly Agree) for usability. The system also strengthened student accountability, parental awareness, and school safety.
3. QR CODE: "Automated Daily Attendance Monitoring System Using QR (Quick Response) Code"	Garrido, C. R. (2024).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Classroom Management. **Context:** The study was conducted among 30 Grade 12 HUMSS students at Tagkawayan National High School during School Year 2022–2023. It focused on using QR code technology to automate daily attendance monitoring and improve the efficiency and accuracy of attendance recording.	**Problem:** Absenteeism remains a significant concern in schools. Traditional attendance monitoring also requires teachers to spend time recording and tracking student attendance. **Action/Solution:** The researcher utilized a QR-code-based attendance monitoring system to record and monitor the attendance of Grade 12 HUMSS students. A quantitative descriptive research design was employed, using a web-based

			survey to evaluate the system's effectiveness. **Outcome:** The findings revealed a significant difference in student attendance before and after implementing the QR-code monitoring system. The system was found to be more effective in monitoring attendance, helping reduce teachers' workload and improve attendance tracking. The study also reported that improved attendance could contribute to better academic performance.
4. Enhancing Attendance Tracking Efficiency and Effectiveness through the Implementation of a QR Code-Based System	Benesa, A. B., Tubice, R. M. A., & Tubice, E. D. T. (2024).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Classroom Management. **Context:** The study was conducted at Abas National High School in Sallapadan, Abra. It involved 124 respondents, including students from Grades 7–12, teachers, and school personnel. It focused on integrating QR code technology into attendance monitoring to improve recording efficiency, accuracy, convenience, and student accountability.	**Problem:** Traditional manual attendance monitoring was time-consuming, prone to human errors, lacked real-time attendance data, and increased teachers' administrative workload. **Action/Solution:** he researchers introduced a QR-code-based attendance monitoring system using personalized QR codes embedded in student identification cards. The system utilized the SAMS application to record and monitor attendance. Surveys and focus group discussions were used to evaluate its effectiveness and gather feedback from students, teachers, and staff. **Outcome:** The findings indicated that QR code technology improved

			attendance tracking efficiency, accuracy, and convenience. Respondents rated faster attendance processing at 3.66, improved accuracy at 3.75, and convenience at 3.77. However, the study also identified occasional technical difficulties and adaptation challenges. Participants expressed support for combining QR-code technology with traditional attendance methods.
5. Decryption of Attendance Monitoring of Senior High School Students at ACTEC Using Zebra Crossing Algorithm	Fabro, B. C., & Anuncio, H. F. (2024).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Classroom Management. **Context:** The study was conducted at Asian Caregiving and Technology Education Centers, Inc. (ACTEC). It focused on developing an online QR-code-based attendance monitoring system for Senior High School students using the Zebra Crossing (ZXing) algorithm to improve attendance recording efficiency, accuracy, and data security.	**Problem:** Traditional attendance monitoring through roll calls and paper-based recording was time-consuming and prone to recording errors. The absence of security measures also created opportunities for unauthorized individuals or students to manipulate attendance records or record attendance remotely. **Action/Solution:** The researchers developed an online attendance monitoring system using QR codes and the Zebra Crossing (ZXing) algorithm. The system incorporated a two-layer security mechanism consisting of verification codes and location-based access to ensure that only authorized students physically present at the school could record their attendance.

			Outcome: The system improved attendance reliability, record accuracy, and administrative efficiency. The security features helped prevent unauthorized attendance logging and remote attendance manipulation while strengthening data protection and integrity.
6. Developing an E-College Monitoring System as a Web-Based Monitoring Tool Application	Mat Nawi, N. A. M., Sapiai, N. S., Mohd Ghazali, S. A., Mat Rusok, N. H., Zulkifli, F. Z., & Mazlan, F. M. (2022).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Student Monitoring. **Context:** The study was conducted at Universiti Teknologi MARA (UiTM), Machang Campus, Malaysia. It focused on developing an E-College Monitoring System to monitor students' entry into and exit from campus, particularly those living in university accommodations. The system aimed to improve student monitoring, outing management, and communication between students, resident staff, and parents.	**Problem:** The existing manual student outing monitoring system was unreliable, time-consuming, and prone to human errors. Resident staff experienced difficulties tracking students' entry and exit, identifying students who had not returned on time, and maintaining accurate records. Manual records also presented risks of data loss and inefficient information management. **Action/Solution:** The researchers developed a web-based E-College Monitoring System using the System Development Life Cycle (SDLC) methodology. The system incorporated student registration, QR-code scanning for check-in and check-out, student outing history, monitoring, reporting, and email notifications to parents and resident staff. **Outcome:** The developed system

			provided a more organized and efficient approach to monitoring student outings. It enabled resident staff to view students' entry and exit records, check timestamps, monitor students who had not returned, and generate reports. The study's qualitative findings indicated that the existing manual system was unreliable and that the proposed technology-based system could improve student monitoring and management. However, the study did not report quantitative measurements of the system's effectiveness.
7. eSAM: Attendance System Using QR Codes in Romblon State University-Cajidiocan Campus	Bacuna, R. D., & Dadiz, B. G. (2022).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Attendance Monitoring. **Context:** The study focused on developing an electronic Student Attendance Management (eSAM) system at Romblon State University–Cajidiocan Campus. It utilized QR codes and mobile devices to automate attendance recording, generate attendance reports, and monitor students' time-in and time-out information.	**Problem:** Manual attendance monitoring was time-consuming and required instructors to spend valuable class time recording student attendance. It also created difficulties in organizing attendance records and generating reports, particularly for large classes. **Action/Solution:** The researchers proposed an electronic attendance monitoring system using QR codes, mobile devices, and a centralized server. Students scanned QR codes containing course, section, subject, date, and IN/OUT information. The system also required students to submit

			facial photographs for identity documentation and provided email notifications confirming successful attendance. **Outcome:** The system enabled instructors and administrators to generate daily and monthly attendance summaries containing students' attendance status, dates, time-in and time-out records, and the number of present, absent, and late students. Reports could also be exported into Excel or PDF formats. The system provided an electronic alternative to paper-based attendance recording and improved attendance record management.
8. Utilizing QR Codes for Smart and Low-cost Student Attendance Acquisition and Monitoring System in Eastern Visayas State University, Philippines	Ebin, J. D., Arandia, P. M., Briso, J. M., Reas, R., Bandala, A., Concepcion, R., Relano, R.-J., Francisco, K., Mayol, A. P., Vicerra, R. R., & Dadios, E. P. (2022).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Attendance Monitoring. **Context:** The study was conducted at Eastern Visayas State University (EVSU), Philippines, involving 10 Bachelor of Science in Electronics and Communications Engineering (BSECE) students. It focused on developing a smart and low-cost attendance acquisition and monitoring system using QR codes and Raspberry Pi technology to improve attendance recording and monitor student absenteeism.	**Problem:** Student absenteeism negatively affects academic achievement. Traditional attendance monitoring methods may also limit the efficiency of recording, storing, retrieving, and monitoring attendance information. **Action/Solution:** The researchers developed a Student Attendance Acquisition System (SAAS) prototype using QR code technology and Raspberry Pi. They also established a web-based Attendance Monitoring System (AMS) capable of storing, retrieving, and displaying attendance

			records. The system was tested with 10 BSECE students over five consecutive days. **Outcome:** Testing revealed no operational problems during the five-day implementation. The prototype demonstrated high efficiency and reliability and showed potential for improving attendance monitoring, identifying frequently absent students, reducing absenteeism, and supporting better academic performance.
9. Efficacy of Scan Attendance Manager Application Using Quick Response Code in Dagohoy National High School, Bohol, Philippines	Galgo, J. J. (2020).	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology and Classroom Management. **Context:** The study was conducted at Dagohoy National High School, Bohol, Philippines, during School Year 2019–2020. It examined the effectiveness of the Scan Attendance Manager (SAM) application using QR codes to monitor student attendance, improve recording efficiency, and provide real-time attendance information to teachers and parents.	**Problem:** Traditional paper-and-pen attendance monitoring was time-consuming and less efficient in tracking student attendance, identifying absentees, and generating reports. Teachers also needed a more reliable system for monitoring attendance and informing parents about their children's attendance status. **Action/Solution:** The researcher implemented the Scan Attendance Manager application using QR codes embedded in students' identification cards. Teachers scanned the QR codes using smartphones to record attendance. The application automatically generated attendance reports and sent SMS

			notifications to parents regarding students' attendance. **Outcome:** The findings showed a significant improvement in class attendance after implementing the application. The average attendance percentage increased from 89.22% before implementation to 99.69% afterward. Teachers also rated the application positively in terms of reliability, efficiency, accuracy, usability, and security, with an overall weighted mean of 3.54 (Strongly Agree). The application improved attendance monitoring, simplified report generation, and strengthened communication between teachers and parents.
10. Campus Entry and Exit Monitoring System	Jacaban, M. J. G., Abrena, M. G., Adalla, J. L., Balangue, M. F., & Orinion, S. M. M. (2023). Campus entry and exit monitoring system [Unpublished undergraduate thesis]. Don Mariano Marcos Memorial State University–South La Union Campus.	**Thematic Area:** Teaching and Learning. **Focus:** Educational Technology, Student Safety, and Student Monitoring. **Context:** The study was conducted at Don Mariano Marcos Memorial State University–South La Union Campus (DMMMSU-SLUC), Agoo, La Union. It focused on developing a QR-code-based campus entry and exit monitoring system to record student arrival and departure, notify parents through SMS, and promote student safety.	**Problem:** The existing manual entry and exit monitoring process required considerable time and effort from students, teachers, campus administrators, and parents. It also made timely and accurate monitoring of student arrivals and departures difficult. **Action/Solution:** The researchers developed a computerized Campus Entry and Exit Monitoring System using QR codes and Visual Basic .NET 2017. Students scanned their QR

			codes when entering and leaving the campus. The system recorded timestamps and student information, sent SMS notifications to parents regarding student arrivals and departures, and incorporated a thermal scanner to check students' temperatures upon arrival. **Outcome:** The evaluation showed favorable results across all assessment criteria. Respondents indicated that the system met user requirements and was usable and efficient. The system provided an organized approach to recording student entry and exit transactions and improved communication with parents.

ACTIVITY 2: Read and analyze the guide questions below. Write your answers in column B.

Guide Questions:	
1. What did you notice about the titles of the action research? What are the common problems to be solved?	We noticed that most of the action research titles focus on using QR code technology and other digital tools to improve attendance monitoring, student tracking, and record management in schools. Even though the studies were conducted in different schools, they all addressed similar concerns. Many schools still rely on manual attendance checking and separate logging systems, which can be time-consuming, prone to errors, and difficult to manage. Another common problem is the challenge of monitoring students' movements within the campus and maintaining accurate records. These studies were conducted to find better and more efficient ways of handling these school-related tasks through technology.

2. What are the common solutions proposed in the problem?	The common solution proposed in most of the studies is the use of QR-code-based systems and mobile applications. These systems help automate attendance monitoring and student tracking, making the process faster, more accurate, and more organized. Instead of manually recording attendance, teachers can simply scan QR codes to generate real-time records. This reduces paperwork, minimizes errors, and saves valuable time for both teachers and school administrators. Similar to these studies, our research proposes a QR-Code-Based Mobile Application that combines attendance recording and exit-and-return logging into one system to make student monitoring more efficient.
3. What do you think about the intention of the teacher-researchers in conducting the research?	We believe that the teacher-researchers conducted these studies because they wanted to improve existing practices in their schools and provide solutions to problems they personally observed. Rather than simply accepting the challenges they encountered, they took the initiative to find practical and effective ways to address them. Their goal was to improve attendance monitoring, strengthen student accountability, lessen teachers' administrative workload, and create a safer and more organized school environment. These studies show that teachers can play an important role in developing innovations that benefit both learners and the entire school community.
4. What significant ideas or concepts did you learn from the activity?	From this activity, we learned that technology can be a powerful tool in solving common problems in schools. We realized that even simple innovations, such as QR codes and mobile applications, can greatly improve efficiency and accuracy in managing student records. We also learned that action research is not only about conducting studies but also about identifying real problems and finding practical solutions that can improve educational practices. Most importantly, we learned that teachers can use research as a way to continuously improve their teaching, classroom management, and the overall learning environment for students.
5. What do you think would be the benefit of conducting action research in your career as future educator?	We believe that conducting action research will be very beneficial in our future careers as educators because it will help us become more reflective, resourceful, and effective teachers. Through action research, we can identify challenges in our classrooms, develop appropriate interventions, and evaluate whether those interventions are successful. It allows us to make decisions based on evidence rather than assumptions. Also, action research encourages continuous learning and professional growth, helping

	us improve our teaching strategies and classroom practices. As future educators, it will enable us to better meet the needs of our students and contribute to the ongoing improvement of educational quality.